Managing Applications

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SWPP Practice Session

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Program Location

- You need program's location to call it
- There are 2 ways to provide program location
  - Canonical path (starting from `/`)
  - Relative path (starting from `./`)

Program Location

- Or, you can omit the parent directories if it is in `$PATH`
  - Commands like sudo, apt, cd are all in `/usr/bin`, which is in `$PATH`
- There are 2 ways to add your program to `$PATH`
  - Add parent directory to the `$PATH` (not recommended)
  - Or create a symbolic link to the program in `/usr/local/bin`

Symbolic Link

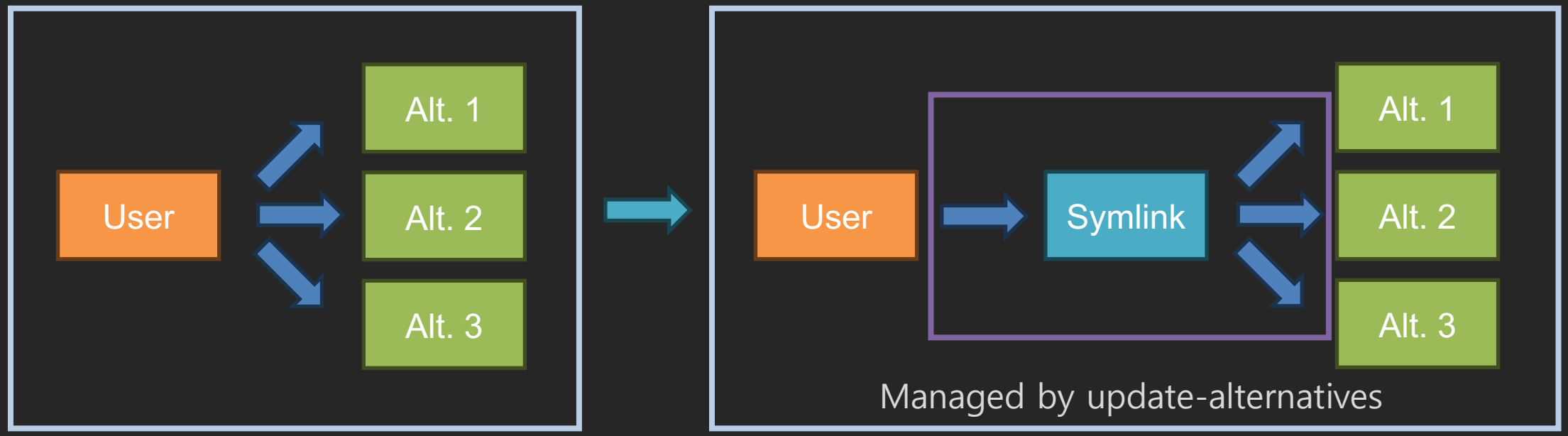
- A pointer to a file
- You can access the file via symlink like it's an actual file
- Deleting symlink does not affect the original file
  - It is just a pointer!
- `ln -s <path> <link-name>`
  - Create link at current directory

Symbolic Link

- In *nix, directories are also files!
- You can access the directory via symlink as well
- Instead of copying the entire directory, try adding symlink
  - Unless you need a separate copy of that directory

update-alternatives

- System utility that uses symlink in the behind
- Adds indirection for multiple versions of applications



update-alternatives

- `sudo update-alternatives --<command>`
  - `--install <config-symlink> <config-name> <alt-path> <prio>`
    - Adds a new indirection with priority (higher means higher priority)
  - `--config <config-name>`
    - Switch between added indirections

Caveats

- This method overrides the default application!
  - Most programs expect distro-default application when running
  - Changing this default may lead to crash or surprising outputs
- Prefer project-specific & temporary solutions
  - E.g. Specify the latest clang++ in a project-local config